

STRUCTURED BILINEAR DIRECTIVITY CONTROL FOR MULTI-CHANNEL PARAMETRIC ARRAY LOUDSPEAKERS

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ABSTRACT

Accurate directivity control of a multi-channel parametric array loudspeaker requires modeling the nonlinear interaction between the carrier and sideband fields. A direct numerical construction of the resulting directivity matrix involves N^2 channel-pair responses for an N -channel array, making model-based optimization computationally expensive. This paper derives a bilinear directivity model that explicitly relates the far-field audio response to the carrier and sideband weight vectors. For identical channels uniformly translated along the array, each channel-pair response is decomposed into an absolute-position phase and a relative-displacement kernel. This decomposition reveals a diagonal–Toeplitz structure in the directivity matrix. By further exploiting angular mirror symmetry and the observed decay of widely separated channel-pair contributions, the number of independently evaluated responses is reduced from N^2 to $L + 1$, where L is the retained maximum channel separation. The structured model is incorporated into an alternating pressure-matching algorithm for joint carrier–sideband control. Numerical results show that the reduced model accurately reconstructs the full-model directivity and retains comparable control performance with substantially fewer independently evaluated channel-pair responses.

Index Terms— parametric array loudspeaker, multi-channel system, directivity control, bilinear model, Toeplitz structure, pressure matching

1. INTRODUCTION

Parametric array loudspeakers (PALs) generate highly directional audible sound through the nonlinear interaction of ultrasonic primary waves [1, 2, 3, 4]. With independently driven channels, a multi-channel PAL can electronically steer the audio beam [5, 6, 7], supporting multibeam reproduction [8], active noise control [9], and sound-zone control [10, 11]. Previous studies have investigated sidelobe suppression [12], beam-pattern synthesis [13], and enhanced beamsteering [14]. Unlike conventional arrays, however, PALs radiate audible sound from a distributed virtual source formed by the carrier and sideband fields [15]. The audio response therefore depends jointly on both sets of channel weights, requiring

nonlinear modeling and control [16].

The modified convolution directivity model (MCDM) efficiently approximates PAL far-field directivity [17] and has been applied to multi-channel beamsteering [14]. Its convolution formulation avoids explicitly constructing the distributed virtual source. Alternatively, wave-expansion methods [18, 19, 20] and numerical approaches [21, 22] provide physics-based predictions of nonlinear sound generation. Nevertheless, constructing a channel-pair response model for directivity optimization can remain computationally demanding. For N channels, the carrier and sideband fields produce N^2 interactions, each contributing a direction-dependent complex response. Independently evaluating all these responses overlooks the spatial redundancy of regularly arranged channels.

This paper exploits this redundancy through a structured bilinear directivity model. Starting from the virtual-source formulation, the audio directivity is expressed in terms of the carrier and sideband weight vectors. For identical, uniformly translated channels, each channel-pair response separates into an absolute-position phase and a kernel describing the nonlinear interaction at a given relative displacement. The resulting diagonal–Toeplitz structure reduces the required responses from N^2 to $2N - 1$ kernels. For mirror-symmetric channel apertures, angular mirror symmetry further reduces this number to N . A banded approximation, motivated by the numerically observed decay of distant-pair contributions, retains only $L + 1$ independent kernels. The structured model is incorporated into an alternating pressure-matching algorithm for joint carrier–sideband control. Numerical results show that the reduced model closely reproduces the full-model directivity and retains comparable control performance with substantially fewer independently evaluated responses.

2. BILINEAR DIRECTIVITY MODEL

2.1. Multichannel PAL Directivity

Consider an N -channel PAL driven by a carrier at angular frequency ω_c and an upper sideband at $\omega_s = \omega_c + \omega_a$. Let $p_{\nu,n}(\mathbf{r}_v, \omega_\nu)$ denote the ultrasonic pressure generated by the n th channel under unit excitation, where $\nu \in \{c, s\}$. The corresponding multichannel transfer vector is

$$\mathbf{p}_\nu(\mathbf{r}_v, \omega_\nu) = [p_{\nu,1}(\mathbf{r}_v, \omega_\nu) \quad \cdots \quad p_{\nu,N}(\mathbf{r}_v, \omega_\nu)]^T. \quad (1)$$

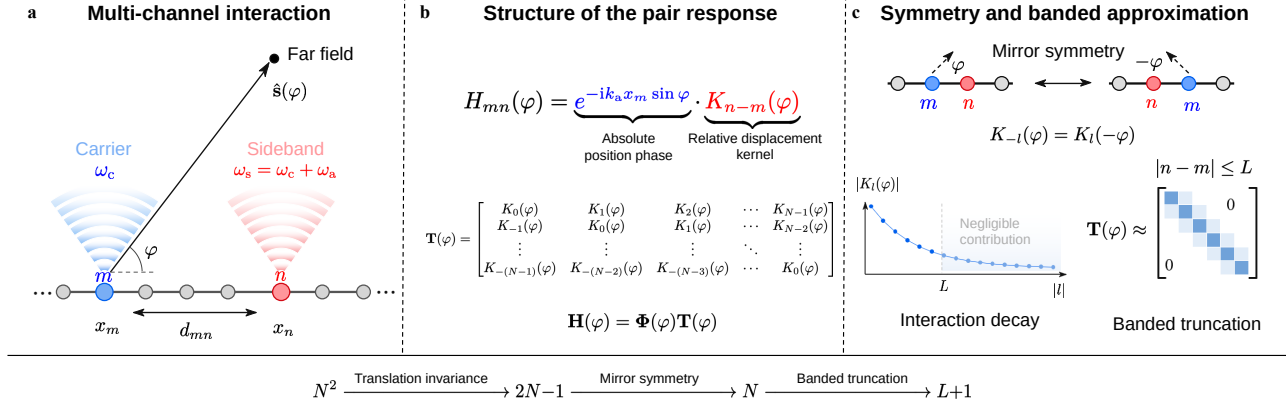


Fig. 1. Bilinear channel interaction and the resulting structured directivity model: (a) multi-channel interaction; (b) diagonal-Toeplitz structure; (c) mirror symmetry and banded approximation.

For a channel-weight vector $\mathbf{w}_\nu$, the total ultrasonic pressure is

$$p_\nu(\mathbf{r}_\nu, \omega_\nu) = \mathbf{p}_\nu^T(\mathbf{r}_\nu, \omega_\nu) \mathbf{w}_\nu. \quad (2)$$

Under the quasilinear approximation, the virtual source at the positive difference frequency ω_a is

$$q_a(\mathbf{r}_\nu, \omega_a) = \frac{\beta \omega_a}{i \rho_0^2 c_0^4} p_c^*(\mathbf{r}_\nu, \omega_c) p_s(\mathbf{r}_\nu, \omega_s). \quad (3)$$

Substituting (2) into (3) gives

$$q_a(\mathbf{r}_\nu, \omega_a) = \frac{\beta \omega_a}{i \rho_0^2 c_0^4} \mathbf{w}_c^H \mathbf{P}_c^*(\mathbf{r}_\nu, \omega_c) \mathbf{P}_s^T(\mathbf{r}_\nu, \omega_s) \mathbf{w}_s. \quad (4)$$

The outer product $\mathbf{P}_c^* \mathbf{P}_s^T$ contains all carrier-sideband channel interactions. Its (m, n) th entry is the local interaction between carrier channel m and sideband channel n .

Following the angular convention in our previous work, the observation direction is parameterized as

$$\hat{\mathbf{s}}(\theta_0, \varphi) = [\sin \varphi \cos \theta_0 \quad \sin \varphi \sin \theta_0 \quad \cos \varphi]^T. \quad (5)$$

This work considers the xOz plane, for which $\theta_0 = 0$ and $\hat{\mathbf{s}}(\varphi) = [\sin \varphi, 0, \cos \varphi]^T$. With $k_a = \omega_a/c_0$, the far-field audio directivity is

$$D_a(\varphi, \omega_a) = -\frac{i \rho_0 \omega_a}{4\pi} \iint \int_{\mathbb{R}^3} q_a(\mathbf{r}_\nu, \omega_a) \times e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_\nu} d^3 \mathbf{r}_\nu. \quad (6)$$

Using (4) in (6) and moving the excitation weights outside the integral yields

$$D_a(\varphi, \omega_a) = \mathbf{w}_c^H \mathbf{H}(\varphi; \omega_c, \omega_s) \mathbf{w}_s, \quad (7)$$

where

$$\mathbf{H}(\varphi; \omega_c, \omega_s) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iint \int_{\mathbb{R}^3} \mathbf{P}_c^*(\mathbf{r}_\nu, \omega_c) \times \mathbf{P}_s^T(\mathbf{r}_\nu, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_\nu} d^3 \mathbf{r}_\nu. \quad (8)$$

The (m, n) th entry of (8) is

$$H_{mn}(\varphi) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iint \int_{\mathbb{R}^3} p_{c,m}^*(\mathbf{r}_\nu, \omega_c) \times p_{s,n}(\mathbf{r}_\nu, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_\nu} d^3 \mathbf{r}_\nu. \quad (9)$$

Therefore, $H_{mn}(\varphi)$ is the complex far-field response generated by one carrier-sideband channel pair. The matrix $\mathbf{H}(\varphi)$ contains the ultrasonic propagation and nonlinear interaction physics, whereas $\mathbf{w}_c$ and $\mathbf{w}_s$ determine how the pairwise responses are combined.

2.2. Displacement-Kernel Structure

Assume that the channels are identical and uniformly arranged along the x axis, with $x_n = (n-1)d$. In a homogeneous propagation environment, the channel fields satisfy

$$p_{\nu,n}(\mathbf{r}_\nu, \omega_\nu) = p_{\nu,0}(\mathbf{r}_\nu - x_n \mathbf{e}_x, \omega_\nu), \quad (10)$$

where $p_{\nu,0}$ denotes the field generated by a reference channel located at the origin.

Substituting (10) into (9) gives

$$H_{mn}(\varphi) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iint \int_{\mathbb{R}^3} p_{c,0}^*(\mathbf{r}_\nu - x_m \mathbf{e}_x, \omega_c) \times p_{s,0}(\mathbf{r}_\nu - x_n \mathbf{e}_x, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_\nu} d^3 \mathbf{r}_\nu. \quad (11)$$

Applying the change of variables $\mathbf{u} = \mathbf{r}_\nu - x_m \mathbf{e}_x$ gives

$$H_{mn}(\varphi) = e^{-ik_a x_m \sin \varphi} \left[-\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iint \int_{\mathbb{R}^3} p_{c,0}^*(\mathbf{u}, \omega_c) \times p_{s,0}(\mathbf{u} - (x_n - x_m) \mathbf{e}_x, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{u}} d^3 \mathbf{u} \right]. \quad (12)$$

Since $x_n - x_m = (n-m)d$, define the displacement kernel

$$K_l(\varphi) = -\frac{\beta \omega_a^2}{4\pi \rho_0 c_0^4} \iint \int_{\mathbb{R}^3} p_{c,0}^*(\mathbf{r}_\nu, \omega_c) \times p_{s,0}(\mathbf{r}_\nu - l d \mathbf{e}_x, \omega_s) e^{-ik_a \hat{\mathbf{s}}^T(\varphi) \mathbf{r}_\nu} d^3 \mathbf{r}_\nu, \quad (13)$$

where $l = n - m$. Equation (12) then reduces to

$$H_{mn}(\varphi) = e^{-ik_a x_m \sin \varphi} K_{n-m}(\varphi). \quad (14)$$

The displacement kernel $K_l(\varphi)$ is the audio directivity generated by a reference carrier channel interacting with a sideband channel displaced by ld . It contains the element radiation, ultrasonic propagation, spatial overlap, and nonlinear interaction associated with that relative displacement. In contrast, the exponential factor in (14) accounts only for the absolute position of the channel pair.

Let $\Phi(\varphi)$ be the diagonal phase matrix whose m th diagonal entry is $[\Phi(\varphi)]_{mm} = e^{-ik_a x_m \sin \varphi}$. Equation (14) can then be written as

$$\mathbf{H}(\varphi) = \Phi(\varphi)\mathbf{T}(\varphi), \quad (15)$$

where

$$\mathbf{T}(\varphi) = \begin{bmatrix} K_0(\varphi) & K_1(\varphi) & \cdots & K_{N-1}(\varphi) \\ K_{-1}(\varphi) & K_0(\varphi) & \cdots & K_{N-2}(\varphi) \\ \vdots & \vdots & \ddots & \vdots \\ K_{-(N-1)}(\varphi) & K_{-(N-2)}(\varphi) & \cdots & K_0(\varphi) \end{bmatrix}. \quad (16)$$

Because each diagonal of $\mathbf{T}(\varphi)$ corresponds to a fixed relative channel displacement, $\mathbf{T}(\varphi)$ is Toeplitz. Therefore, the complete N^2 channel-pair responses can be constructed from only $2N - 1$ displacement kernels, $K_l(\varphi)$ for $l = -(N - 1), \dots, N - 1$.

2.3. Mirror Symmetry and Banded Approximation

Let $\mathbf{R} = \text{diag}(-1, 1, 1)$ denote reflection about the yOz plane. If the reference channel and propagation environment are mirror-symmetric, then $p_{\nu,0}(\mathbf{R}\mathbf{r}_v, \omega_\nu) = p_{\nu,0}(\mathbf{r}_v, \omega_\nu)$. Moreover, $\hat{\mathbf{s}}(-\varphi) = \mathbf{R}\hat{\mathbf{s}}(\varphi)$. Applying the transformation $\mathbf{r}_v \leftarrow \mathbf{R}\mathbf{r}_v$ to (13) yields

$$K_{-l}(\varphi) = K_l(-\varphi), \quad l = 1, \dots, N - 1. \quad (17)$$

Thus, the negative-displacement kernels can be reconstructed from the N nonnegative-displacement kernels. Equation (17) is an angular mirror relation rather than a complex-conjugate symmetry.

The integrand in (13) contains the spatial overlap between two ultrasonic fields separated by $|l|d$. This overlap is expected to decrease with channel separation, although its actual decay depends on the array geometry and must be verified numerically. Motivated by this behavior, define the banded approximation

$$[\mathbf{H}_L(\varphi)]_{mn} = \begin{cases} H_{mn}(\varphi), & |n - m| \leq L, \\ 0, & |n - m| > L. \end{cases} \quad (18)$$

The corresponding audio directivity is approximated by

$$D_a(\varphi) \approx \mathbf{w}_c^H \mathbf{H}_L(\varphi) \mathbf{w}_s. \quad (19)$$

Algorithm 1 Alternating Joint Directivity Control

Require: $\{\mathbf{H}_L(\varphi_i), d(\varphi_i), \alpha_i\}_{i=1}^M$

- 1: Initialize $\mathbf{w}_c^{(0)}$ and $\mathbf{w}_s^{(0)}$
 - 2: **repeat**
 - 3: $\mathbf{w}_s^{(t+1)} \leftarrow \arg \min_{\mathbf{w}_s} J(\mathbf{w}_c^{(t)}, \mathbf{w}_s)$
 - 4: $\mathbf{w}_c^{(t+1)} \leftarrow \arg \min_{\mathbf{w}_c} J(\mathbf{w}_c, \mathbf{w}_s^{(t+1)})$
 - 5: $t \leftarrow t + 1$
 - 6: **until** convergence
 - 7: **return** $\mathbf{w}_c^*, \mathbf{w}_s^*$
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Translation invariance, mirror symmetry, and banded truncation therefore reduce the number of independently evaluated channel-pair responses according to

$$N^2 \longrightarrow 2N - 1 \longrightarrow N \longrightarrow L + 1. \quad (20)$$

The derivation above uses integration over $\mathbb{R}^3$, for which spatial translation does not change the integration domain. In the numerical implementation, the integral is evaluated over a finite domain Ω , which must be sufficiently large to contain the translated virtual-source distributions.

3. JOINT DIRECTIVITY CONTROL

Given the desired complex directivity $d(\varphi_i)$ at M observation angles, the carrier and sideband weights are obtained from

$$\min_{\mathbf{w}_c, \mathbf{w}_s} \sum_{i=1}^M \alpha_i |\mathbf{w}_c^H \mathbf{H}_L(\varphi_i) \mathbf{w}_s - d(\varphi_i)|^2 + \lambda_c \|\mathbf{w}_c\|_2^2 + \lambda_s \|\mathbf{w}_s\|_2^2. \quad (21)$$

Problem (21) is nonconvex with respect to the two weight vectors jointly. When one weight vector is fixed, however, the predicted directivity becomes linear in the other, and the corresponding subproblem is an unconstrained complex-valued regularized least-squares problem. Each subproblem is therefore solved in closed form, and the carrier and sideband weights are updated alternately until convergence, as summarized in Algorithm 1.

4. NUMERICAL RESULTS

4.1. Simulation Setup

A 24×24 PAL composed of circular transducers with radius 5 mm is considered. The transducers are closely packed without staggering at a spacing of $d = 10$ mm. Each column is driven as one channel, resulting in $N = 24$ channels. The carrier and upper-sideband frequencies are 40 and 41 kHz, respectively, producing a 1 kHz audio signal.

The directivity is evaluated over $\varphi \in [-80^\circ, 80^\circ]$ with a 1° interval. The virtual-source domain is $30 \times 1.4 \times 24$

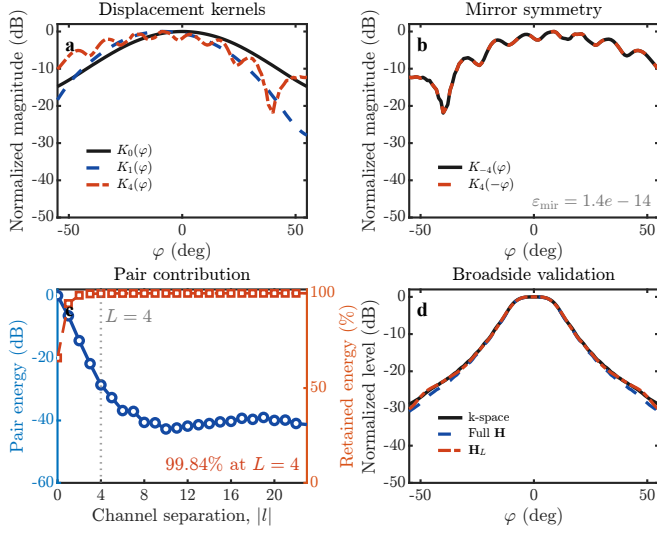


Fig. 2. Structured-model validation: (a) displacement kernels; (b) mirror reconstruction; (c) pair energy and cumulative retained energy; (d) broadside directivities from k-space simulation and the full and banded models.

m. The desired directivity is centered at $\varphi_0 = 20^\circ$, with a passband of $\pm 8^\circ$ and a raised-cosine transition extending to $\pm 20^\circ$. The main-lobe and suppression-region weights are 20 and 10, respectively, and the regularization parameter is 0.3.

4.2. Structured-Model Validation

Figure 2(a) shows that the displacement kernels have separation-dependent angular responses. Figure 2(b) verifies the mirror relation $K_{-l}(\varphi) = K_l(-\varphi)$. The relative errors are 3.89×10^{-15} and 1.41×10^{-14} for $l = 1$ and $l = 4$, respectively.

For $\mathcal{P}_l = \{(m, n) : |n - m| = l\}$, the mean pair energy is $E_l = |\mathcal{P}_l|^{-1} \sum_{(m,n) \in \mathcal{P}_l} \sum_{i=1}^M |H_{mn}(\varphi_i)|^2$. The cumulative retained energy is $\eta_L = \sum_{l=0}^L |\mathcal{P}_l| E_l / \sum_{l=0}^{N-1} |\mathcal{P}_l| E_l$.

As shown in Fig. 2(c), $L = 4$ retains 99.84% of the total energy. Translation invariance, mirror symmetry, and banded truncation therefore reduce the required responses from $N^2 = 576$ to $L + 1 = 5$. Figure 2(d) shows close agreement among direct k-space simulation, the full model, and the banded model, with a full-to-banded RMSE of 1.22 dB.

4.3. Joint Directivity Control

Figure 3(a) compares conventional steering, sideband-only PM, and joint PM. Their maximum suppression-region levels are -8.40 , -10.08 , and -14.14 dB, respectively. Sideband-only PM shifts the peak to 23° , whereas joint PM maintains

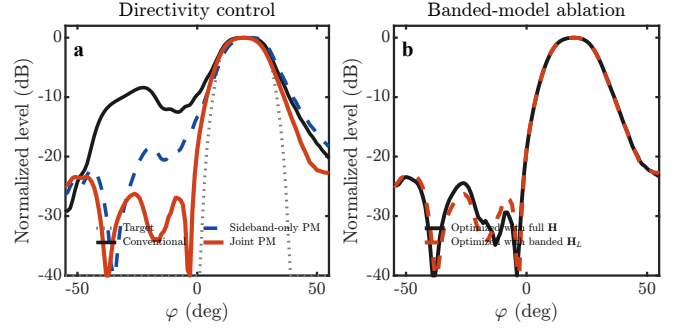


Fig. 3. Directivity control at 20° : (a) target and responses from conventional steering, sideband-only PM, and joint PM; (b) joint PM using full and banded models. All responses are evaluated using the full matrix.

the desired 20° direction. Their normalized target errors are 0.374, 0.210, and 0.125, respectively.

Figure 3(b) compares joint PM optimized using the full and banded matrices, with both responses evaluated using the full matrix. Their suppression levels are -14.09 and -14.14 dB, and their target errors are 0.123 and 0.125, respectively. Thus, the five-kernel model preserves the performance of full-matrix optimization.

5. CONCLUSION

A structured bilinear model was developed for multi-channel PAL directivity control. Translation invariance and mirror symmetry reduce the required channel-pair responses from N^2 to N , while banded truncation further reduces them to $L + 1$. For the considered 24-channel PAL, five kernels retain 99.84% of the matrix energy and reproduce the full-model response. Joint carrier-sideband PM reduces the undesired-lobe level by 5.74 dB over conventional steering while maintaining the desired beam direction.

6. COMPLIANCE WITH ETHICAL STANDARDS

No ethical approval was required for this numerical simulation study. The authors have no relevant financial or non-financial interests to disclose.

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